

# Yunwan Fu

+1 323-922-0845 | +86 133-9447-4762 | [asterrose@sjtu.edu.cn](mailto:asterrose@sjtu.edu.cn) | [Homepage](#)

## EDUCATION

### Shanghai Jiao Tong University

*Bachelor of Engineering in Automation (IEEE Pilot Program)*

Shanghai, China

Sep 2024 – Jun 2028

**Overall Academic Score:** 97/100 (**Top 1/25 in Major**)

**Relevant Coursework (All A+ Grades):** C++ Principles, Video and Multimedia Retrieval, Electronic Circuit Laboratory, Linear and Convex Optimization, Linear Algebra, Probability and Statistics, Mobile Robotics

## RESEARCH INTERESTS

Embodied AI, Vision-Language-Action (VLA), Agents, Humanoid, Vision-Language Models (VLMs), and agent harnesses for robotics.

## RESEARCH PROJECTS

### PSI Lab , USC

*Research Intern, Advisor: Prof. Yue Wang*

Los Angeles, USA

Mar 2026 – Present

#### • **PhySA: Physical Smoothness Agent for Humanoid Locomotion**

- \* Designed a dual-channel reward framework decoupling VLM aesthetic scoring from physical smoothness correction, resolving their inherent conflict in humanoid RL.
- \* Achieved 59.5% lower joint acceleration, 43% lower mechanical power, and 57% longer episodes over Prompt2Policy on MuJoCo Humanoid-v5.
- \* Implemented asynchronous VLM scheduling ( $24\times$  fewer calls) and phase-triggered reward editing; manuscript in preparation for CoRL 2026 Workshop.

#### • **$\Psi$ -X: Egocentric Video Annotation via VLM**

- \* Deployed EgoAnno pipeline with Qwen-30B (vLLM) and DINOv2 on EgoDex dataset; curated long-horizon pick-and-place tasks for targeted evaluation; currently running full pipeline to benchmark zero-shot VLM annotation performance.

#### • **H2: Cross-Embodiment Whole-Body Control Transfer for Unitree Humanoid**

- \* Migrated NVIDIA SONIC's whole-body policy from Unitree G1 (29 DoF) to H2 (31 DoF) via name-matched weight transfer, retargeted 131,454 SMPL motions into H2 trajectories (100% coverage, 10.9 mm error), and trained four 120k-step policies in Isaac Lab, identifying jitter and sim-to-sim instability as remaining blockers to real-robot deployment.

## AWARDS & HONORS

A-Level Academic Scholarship, Shanghai Jiao Tong University

Dec 2025

Rayliant-Sequoia Talent Development Fund

Nov 2025

National First Prize, China Undergraduate Mathematical Contest in Modeling

Nov 2025

“Merit Student” of the 2024-2025 Academic Year, Shanghai Jiao Tong University

Oct 2025

Finalist Award, Mathematical Contest in Modeling (MCM/ICM), COMAP

May 2025

## TECHNICAL SKILLS

**Languages:** Chinese (Native), English (Fluent, CET-6 562/710)

**Technical Skills:** Python, C++, LaTeX, Linux, ROS/ROS2, Gazebo, OpenCV, MuJoCo, Isaac Sim, PID.